

Titus Loftin

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EDUCATION

The University of Texas at Austin	Bachelor of Science, Chemical Engineering Minors and Certificates: Computational Science and Engineering, Programming and Computation, and Statistics and Data Science Overall GPA: 3.80	May 2027
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EXPERIENCE

Soft Tissue Biomechanics Lab - Undergraduate Researcher	Sept. 2024 - Present
- Developed a method of identifying heterogeneous soft materials by collecting 8 different types of strain data from 120 different materials, and using neural ODE fields for mechanical characterization - Created an in-vitro setup to run mechanical tests on blood clots in the middle cerebral artery that measures pressure and flow conditions up to an accuracy of 0.1%	
Hill Country bible Church - Facilities Technical Intern	June 2020 - August 2024
- Designed and built portable baptism tubs for use in the church, reducing potential costs by ~\$2K if bought from a third party - Decreased energy use by 15% installing lighting fixtures throughout the facilities - Installed a more efficient pond irrigation system to optimize water use by 8%	

PROJECTS

Smart financial planner
Developed a full-stack budget planner with React, Node.js, and PostgreSQL, featuring personalized spending reduction suggestions.
3D CFD program
Created a 3D CFD simulation tool in Julia to model pulsatile and steady-state fluid flow, establishing scaling laws around complex geometries.
Market information Chrome extension
Built a Chrome extension using JavaScript and Chrome APIs to deliver real-time market updates and portfolio tracking.

SKILLS

Programming Languages: Python, MATLAB, C++, C#, Java, Typescript, Julia

Design Tools: SolidWorks, Fusion360, LabVIEW, OpenFOAM

Manufacturing: Welding, Soldering, CNC Machining

LEADERSHIP EXPERIENCE AND ACTIVITIES

Chemical Engineering Undergraduate Advisory Board - Director and Founder	Fall 2024 - Present
Founded a student organization to improve the CHE department. Recruited ~25% of the Chemical Engineering student body during the 2024-2025 school year.	
Tau Beta Pi - Vice President Corporate Relations	Spring 2025 - Present
Managed relationships with companies Blue Origin and Lockheed Martin, brought in corporate sponsors such as CDM Smith.	
Longhorn Developers - General Member	Fall 2024 - Present
Member for Longhorn Developers, a student organization at UT who works on different projects to improve some of the technology at the University. One of the notable projects is UT Registration Plus, a Chrome extension that has allowed 60,000 students to create their schedules.	